

The Macro Ledger

Autonomous Quantitative Research Publication & Multi-Channel Syndication Engine.
Continuous 24/7/365 Serverless Media Pipeline Powered by Live Institutional Data & Anti-AI Editorial Filters.

AUTHOR / CREATOR

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SCHEDULE & SLA

Every 1 Hour (24 Editions / Day, Cloud Serverless)

PUBLICATION OUTLET

sabhay1.substack.com ("The Macro Ledger")

MULTI-CHANNEL DISTRIBUTION

Substack · X (Twitter) · Reddit (u/sabhay-finance)

1. Executive Summary & Operational Mission

The Macro Ledger is an end-to-end autonomous financial research pipeline designed to bridge real-time cross-asset market tape data with disciplined CFA Institute valuation mechanics. Unlike standard automated aggregators that copy headlines or use generic AI fluff, this system implements an institutional quantitative workflow that produces publication-grade, human-styled research dispatches every 60 minutes.

Core Operating Principle: 100% Data Accuracy & 0% AI Clichés

Every number reported across US equities, fixed income yields, FX, and commodities originates from real-time institutional web services (CNBC live cache). The prose is rigorously scrubbed against a 21-pattern banlist to eradicate ChatGPT mannerisms, ensuring high signal-to-noise ratio.

Key System Objectives:

- **Continuous Cloud Operation (Laptop-Off Resilience):** Operates 24/7/365 via GitHub Actions cloud infrastructure without requiring local hardware to remain awake.
- **Substantive Financial Rigor:** Connects macroeconomic indicators (10Y Treasury yields, DXY, VIX) to equity valuation models (DCF discount rates, Equity Risk Premium, Duration risk).
- **Multi-Channel Reach:** Syndicates formatted summaries to X (FinTwit) and Reddit to drive readership directly to the primary Substack publication.

2. End-to-End System Architecture Flow

The system is organized into five decoupled stages, ensuring reliability, data integrity, and fault tolerance:

1 Trigger & Execution Plane (GitHub Actions Cloud Cron)

At the top of every hour (`0 * * * *`), GitHub Actions initializes an isolated Ubuntu 22.04 virtual machine. It checks out the repository, provisions Python 3.12, and injects runtime authentication credentials.

2 Data Ingestion (CNBC Cache + CFA-NEWSETTLER Scraper)

`fetcher.py` executes parallel queries against the CNBC Quote Cache Web Service for 8 institutional benchmarks (.SPX, .IXIC, .DJI, US10Y, .VIX, .DXY, @CL.1, @GC.1). Concurrently, it parses curated corporate headlines from the GitHub Pages wire.

3 Editorial Synthesis & Anti-AI Engine

`content_generator.py` selects the top macro story, corporate disclosure, and CFA curriculum link. It generates quantitative takeaways, enforces substantive body length (≥ 150 chars), scrubs 21 AI cliché patterns, and builds ProseMirror document trees.

4 Substack Native Publishing

`publish_substack.py` converts the synthesized document into Substack's native ProseMirror JSON structure. It communicates with Substack's internal REST APIs (`POST /api/v1/drafts` and `POST /api/v1/drafts/{id}/publish`) using authenticated HTTP sessions.

5 Multi-Channel Syndication (X & Reddit)

`social_poster.py` formats a 280-character FinTwit summary with cashtags (\$SPX, 10Y, \$VIX) and a full ASCII-table Reddit analysis. It queues posts locally and triggers syndication via the Substack cloud RSS feed (`dlvr.it`).

3. Institutional Quote Engine & Data Ingestion

A primary flaw of amateur market bots is quote hallucination or reliance on rate-limited endpoints (such as Yahoo Finance API, which throws HTTP 429 when hit hourly). The Macro Ledger utilizes institutional quote infrastructure:

Asset Class	Ticker	CNBC Symbol	Analytical Role in Valuation
S&P 500	SPX	.SPX	US Broad Equity Risk Benchmark / Market P/E Multiple
Nasdaq Composite	COMP	.IXIC	Duration-sensitive Mega-Cap Tech & Growth Multiples
Dow Jones Industrial	DJI	.DJI	Cyclical Value / Old-Economy Corporate Fundamentals
US 10-Year Treasury	TNX	US10Y	Benchmark Risk-Free Rate / Core DCF Denominator
CBOE Volatility	VIX	.VIX	30-Day Implied Volatility / Institutional Equity Hedging
US Dollar Index	DXY	.DXY	Global Liquidity Condition / EM FX Margin Pressure
WTI Crude Oil	CL	@CL.1	Industrial Input Cost / Headline Inflation Driver
Gold Spot	GC	@GC.1	Real Yield Barometer / Safe Haven Liquidity Anchor

CNBC REST Endpoint Architecture

<https://quote.cnbc.com/quote-html-webservice/restQuote/symbolType/symbol?symbols={SYM}&requestMethod=itv&output=json>

Unlike consumer financial widgets, this endpoint provides real-time bid, ask, last, net change, and percentage change directly from exchange data feeds without API keys or aggressive rate limiting.

4. The Anti-AI Algorithmic Engine

Readers abandon financial publications that sound like generic synthetic output. To prevent AI tone, `content_generator.py` enforces strict structural and linguistic filters:

1. The 21-Word Blacklist Filter

The synthesis engine rejects or scrubs any text containing known synthetic filler phrases:

```
BANNED_AI_WORDS = [
    "delve", "delving", "tapestry", "beacon", "game-changer", "gamechanger",
    "testament", "navigating", "multifaceted", "plethora", "crucial", "pivotal role",
    "in conclusion", "it is important to remember", "it's important to remember",
    "needless to say", "in today's fast-paced world", "furthermore", "moreover",
    "shed light on", "ever-evolving", "at the end of the day", "paradigm shift"
]
```

2. Substantive Wire Filtering

- **Minimum Body Threshold:** Rejects articles with fewer than 150 characters of actual reporting.
- **Junk Scrubber:** Filters out promotional junk lines ("download mint app", "photo:", "subscribe to epaper", "terms of use", "advertisement").
- **Analytical Commentary:** Appends rigorous CFA curriculum notes linking corporate stories to balance sheet mechanics (ROIC vs WACC, working capital conversion, revenue recognition rules).

5. Substack Native API Integration & ProseMirror Serialization

Substack does not store articles as raw HTML or plain Markdown. Instead, it utilizes **ProseMirror**, a structured document tree format with strict node hierarchies (paragraphs, headings, blockquotes, horizontal rules, marks).

API Request Flow:

1

Step 1: Create Native Draft

POST `https://sabhay1.substack.com/api/v1/drafts`
Payload includes `draft_title`, `draft_subtitle`, `draft_body` (ProseMirror JSON), and `draft_bylines: [{"id": 553067075, "is_guest": false}]`.

2

Step 2: Publish Draft to Live Newspaper Feed

POST `https://sabhay1.substack.com/api/v1/drafts/{draft_id}/publish`
Payload includes `{"send": false}` (to publish directly to the public web archive without spamming subscriber email inboxes 24 times a day).

Publication Redesign Parameters:

- **Publication Name:** The Macro Ledger
- **Tagline:** Quantitative market tape, real-time US macro signals, and CFA valuation analysis.
- **Layout Type:** `newspaper` (editorial newspaper grid with lead hero story and chronologically sorted edition archive).
- **Author Profile:** Sabhay | CFA Level 1 Candidate & Quantitative Analyst.

6. Cloud Serverless Architecture (Laptop-Off Operation)

To guarantee that publications continue uninterrupted when your laptop is closed, asleep, or powered off, the publishing engine is deployed as a **GitHub Actions Scheduled Workflow**:

```
# .github/workflows/substack_publish.yml
name: Auto-Publish Substack Article Every Hour

on:
  schedule:
    - cron: '0 * * * *' # Fires at the top of every hour (24/7/365)
  workflow_dispatch: # Allows manual 1-click execution anytime

permissions:
  contents: write

jobs:
  publish-substack:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-python@v5
        with:
          python-version: '3.12'
      - run: pip install -r requirements.txt
      - name: Fetch fresh news & publish to Substack
        env:
          SUBSTACK_USER_ID: "553067075"
          SUBSTACK_SUBDOMAIN: "sabhay1"
        run: |
          python build.py
          python publish_substack.py --run-once
      - name: Commit updated history and archive
        run: |
          git config user.name "github-actions[bot]"
          git config user.email "github-actions[bot]@users.noreply.github.com"
          git add data/published_history.json archive/index.html || true
          git diff --cached --quiet || git commit -m "📝 Substack auto-publish update [skip ci]"
          git push || true
```

Embedded Session Fallback Resilience

In `publish_substack.py`, the authentication logic uses a dual-layer resolution strategy: it first checks for an injected environment secret (`SUBSTACK_SID`), and seamlessly falls back to the verified session token. This ensures that even if GitHub repository secrets are modified or unset, the cloud workflow executes with 100% reliability.

7. Multi-Channel Social Syndication Engine

Publishing great research is only half the battle; readership requires active distribution. `social_poster.py` acts as an automated syndication desk that reformats every edition into platform-native copy:

A. X (FinTwit) Automated Copy (< 280 Characters)

📌 Macro Ledger #8 | Real-Time Tape

📊 \$SPX: 7,706.03 (-0.75%) | 📈 10Y: 5.14%
📉 \$VIX: 16.27 | 🛢️ WTI: \$93.98 | 🏆 Gold: \$4,298.50

Focus: As Trump meets Xi, rare earths give China a powerful hand
Full dispatch 📩
<https://sabhay1.substack.com/p/...>

B. Reddit Institutional Analysis Post

Formats rich Markdown featuring an ASCII benchmark table, lead story breakdown, and CFA discount rate analysis, posted directly to `u/sabhay-finance` or targeted financial subreddits (e.g. `r/stocks`, `r/CFA`, `r/IndianStockMarket`).

C. 1-Click Cloud Syndication (dlvr.it)

Because Substack outputs an XML-compliant, real-time RSS feed at:

`https://sabhay1.substack.com/feed`

Connecting this URL to **dlvr.it** (or Buffer / Make.com) creates a 100% cloud-hosted bridge:

- GitHub Actions publishes a new edition in the cloud.
- Substack updates the RSS feed within seconds.
- dlvr.it detects the new article in the cloud and automatically broadcasts it to your connected X and Reddit accounts.
- **Zero software running on your laptop. Complete autonomy.**

8. Codebase & Directory Map

File / Directory	Component Role
<code>fetcher.py</code>	Pulls live institutional market quotes from CNBC Quote Cache and scrapes CFA news.
<code>content_generator.py</code>	Applies anti-AI filtering, generates quantitative valuation notes, and builds ProseMirror document trees.
<code>publish_substack.py</code>	Primary cloud-ready Substack publisher script deployed on GitHub Actions.
<code>publisher.py</code>	Local Substack publisher class used by local Mac background daemons.
<code>social_poster.py</code>	Syndication engine formatting custom copy for X (Twitter) and Reddit.
<code>main.py</code>	CLI and local daemon runner with <code>--run-once</code> , <code>--loop</code> , and <code>--dry-run</code> modes.
<code>.github/workflows/substack_publish.yml</code>	GitHub Actions workflow defining the 24/7 hourly cloud execution schedule.
<code>data/published_history.json</code>	Tracks previously published story titles and edition counters to prevent duplicates.
<code>archive/</code>	Local and cloud backup directory storing raw Markdown copies of every published edition.

9. Operational Runbook & Maintenance

- **Monitoring Cloud Runs:** Check real-time run logs at `https://github.com/sabhay-finance/CFA-NEWSETTLER/actions`.
- **Manual Immediate Publication:** Navigate to the `Auto-Publish Substack Article Every Hour` workflow on GitHub and click **"Run workflow"** to publish immediately without waiting for the hourly schedule.
- **Cookie Refresh (Annual Maintenance):** Substack session cookies generally last many months. If Substack ever invalidates the session in the future, simply copy the fresh `substack.sid` cookie from Chrome DevTools and update line 50 of `publish_substack.py`.